

root user can execute the Cron job. On most GNU/Linux distributions, only the `/etc/cron.deny` file exists and it is empty.

So far we have discussed the Cron daemon, the Cron table and the `crontab` utility. Now let us understand the format of the Cron table. In the Cron table, each field is separated by a space; its format is as follows:

```
{minute} {hour} {day of month} {month} {day of week} {absolute
path of command or script}
```

The first five are time and date fields. While parsing the Cron table, blank lines, leading spaces and tab characters are ignored. Also, any line that begins with the hash (#) character is treated as a comment and is not processed.

The allowed values for the time and date fields are given in Table 1.

Table 1

Fields	Allowed values
minute	0 - 59
hour	0 - 23
date	1 - 31
month	1 - 12 (or names); 1 = January and 12 = December.
day of the week	0 - 7 (or names); Sunday is either 0 or 7.

Instead of numeric values, we can also use names for the 'month' and 'day of the week' fields but ranges or lists of names are not allowed. The first three letters of the particular day or month can be used in place of the 'month' or 'day of week' fields. These names are case insensitive.

Additionally, there are eight special strings that we can use as short cuts in the Cron table to specify the time and date (see Table 2).

Table 2

Field	Meaning
@reboot	Run once at system startup
@yearly	Run once a year; same as "0 0 1 1 **"
@annually	Same as @yearly
@monthly	Run once a month; same as "0 0 1 * **"
@weekly	Run once a week; same as "0 0 * * 0"
@daily	Run once a day; same as "0 0 * * *"
@midnight	Same as @daily
@hourly	Run once an hour; same as "0 * * * *"

The command(s) in the Cron table are specified with an absolute path. This is needed because Cron runs in different environments and the 'PATH' environment variable may not be set. `/bin/sh` is treated as the default shell by the Cron daemon.

Several environment variables are set by the Cron daemon at startup. Cron tables are parsed from top to

bottom; hence, environment settings are applicable only to those commands which are specified after setting the environment variables. By default, the SHELL is set to `/bin/sh` and the LOGNAME and HOME environment variables are set from `/etc/passwd` file. LOGNAME is the name of the user executing the Cron job. The default value of PATH is `/usr/bin:/bin`. HOME, PATH and SHELL environment variables can be overridden from the Cron table.

By default, after successful command execution, the output is mailed to the owner of the Cron table. We can override this default behaviour by setting the MAILTO environment variable. If MAILTO is set and is not empty, then the output of the command is mailed to the user named. In MAILTO, multiple recipients can be specified by a comma-separated list. If an empty value is assigned to MAILTO (e.g., MAILTO=""), then no mail is sent.

In the Cron table, we can assign values to environment variables just as in a shell assignment. The simple example below will give a better idea about how an environment variable is used:

```
# Set environment variables.
PATH=/usr/bin:/bin:/opt/additional_packages
SHELL=/bin/bash
MAILTO="jerry@acme.com"
# Check disk usage every week day at 11:30 PM.
30 23 * * 1-5 /home/tom/check_disk_usage.sh
```

The Cron tab utility provides certain operators that we can use to specify multiple values in a field. Table 3 describes operators supported by the Cron tab.

Table 3

Operator	Meaning
Asterisk (*)	Implies that the command should be executed at every instance of time. For example, an asterisk in the minute field means the command should be executed every minute; or an asterisk in the hour field means the command should be executed every hour, and so on.
Comma (,)	Using a comma, we can specify a list of values, e.g., to execute a Cron job four times a day, in the hour field, we can specify 12,15,18,21.
Hyphen (-)	Using a hyphen, we can provide a range of values, e.g., the weekend can be specified as 6-7 in the 'day of the week' field.
Forward slash (/)	This operator represents a particular division of time, e.g., */5 in the minute field is to execute the command every five minutes or */3 in the 'hour' field is to execute the command every three hours.

Along with the operators, the Cron tab also provides certain command line options, which we can use to create or